

- Louis is selling his art at an auction. However, he is afraid no one will show up to bid for his art. He can pay a random person \$5 to show up and bid. Each bidder then bids \$ X , where X is a random variable distributed according to the density

$$f(x) = \begin{cases} a \sin\left(2\pi \frac{x}{1000}\right) + b & 0 < x < 1000 \\ 0 & \text{else} \end{cases}$$

for $a, b \in \mathbb{R}$

- Find possible ranges for a and b .
- Let a, b take their largest possible values. Find the probability p of a bid over \$700, and the expected bid amount.
- Suppose all bidders place their bids independently. Suppose n bidders appear at the auction. Let N denote the number of bidders who bid over \$700
 - * Find and simplify $g(k) = \frac{P(N=1|n=k+1)}{P(N=1|n=k)}$ in terms of p and k .
 - * Find the smallest value of k such that $g(k) < 1$. What does this represent?
- Suppose instead X had an unknown distribution with $\sigma = 70$
 - * There are $n = 45$ bidders present at the auction, and 6 of them bid over \$700. Find a 95% confidence interval for the value of $P(X > 700)$.
 - * ** A survey discovered that out of $n = k$ bidders, the average bid amount was \$456.5. It was previously assumed that the mean bid amount is \$500. Under a significance level of $\alpha = 0.05$, a hypothesis test was carried out for the mean amount.
 - State the null and alternative hypothesis
 - Find a 95% confidence interval for the mean bid amount in terms of n . Suppose $P(-c < Z < c) = 0.95$.
 - Using the above confidence interval, find the value(s) of n for which we accept the null hypothesis
 - It was later found that the true mean was actually \$520. Find the value(s) of n that results in a type 2 error rate of under 0.3
- As auctions usually do, the art will be sold to the highest bidder. Given n bidders, maximum bid X_{max} is distributed according to the density

$$g(x) = \begin{cases} n \frac{x^{n-1}}{1000^n}, & 0 < x < 1000 \\ 0, & \text{else} \end{cases}$$

- * Find Louis's expected gain from this auction in terms of n
 - * How many random people should Louis bribe to join his auction?
- Suppose Louis's art can be resold. The resell value depends on whether he used real gold pigment or not. The probability he used real gold pigment was 0.3. Given real gold pigment was used, the resell value is U . Otherwise, the resell value is V . $U \sim g$ where

$g(x) = \frac{1}{1500}e^{-x/1500}$, $x > 0$ and V is with mass function $p(x)$, where $p(0) = \frac{3}{10}$, $p(400) = p(600) = p(650) = \frac{7}{30}$. Find the expected resell value of his art.

1. Ranges for a and b

- $\int_0^{1000} f(x) dx = 1 \implies b = \frac{1}{1000} = 0.001$
- $f(x) \geq 0 \implies |a| \leq b \implies a \in \left[-\frac{1}{1000}, \frac{1}{1000}\right]$

2. Largest a, b, Probability p, and $E[X]$

- Largest values: $a = \frac{1}{1000}$, $b = \frac{1}{1000}$
- $p = P(X > 700) = \int_{700}^{1000} \left(\frac{1}{1000} \sin\left(\frac{2\pi x}{1000}\right) + \frac{1}{1000}\right) dx = 0.3 - \frac{1 - \cos(1.4\pi)}{2\pi} \approx 0.0917$
- $E[X] = \int_0^{1000} x f(x) dx = 500 - \frac{500}{\pi} \approx 340.85$

3. Expression for $g(k)$

- $N \sim \text{Binomial}(n, p) \implies P(N = 1 | n) = np(1-p)^{n-1}$
- $g(k) = \frac{(k+1)p(1-p)^k}{kp(1-p)^{k-1}} = \frac{k+1}{k}(1-p)$

4. Smallest k such that $g(k) < 1$

- $\frac{k+1}{k}(1-p) < 1 \iff k > \frac{1-p}{p} \implies k = \left\lfloor \frac{1-p}{p} \right\rfloor + 1$
- Represents the sample size k at which $P(N = 1 | n)$ reaches its maximum mode and starts declining

5. 95% Confidence Interval for $P(X > 700)$

- $\hat{p} = \frac{6}{45} = \frac{2}{15} \approx 0.1333$, $SE = \sqrt{\frac{\hat{p}(1-\hat{p})}{45}} \approx 0.0507$
- $CI_{0.95} = \hat{p} \pm 1.96 \cdot SE = [0.0340, 0.2327]$

6. Hypothesis Testing on Mean Bid

- Hypotheses: $H_0 : \mu = 500$ vs. $H_1 : \mu \neq 500$
- 95% CI: $\left[456.5 - c\frac{70}{\sqrt{n}}, 456.5 + c\frac{70}{\sqrt{n}}\right]$ (with $c \approx 1.96$)
- Accept H_0 if $500 \leq 456.5 + 1.96\frac{70}{\sqrt{n}} \implies \sqrt{n} \leq 3.154 \implies n \leq 9$, so $n \in \{1, 2, \dots, 9\}$
- Type II error $\beta < 0.3$ when $\mu = 520$: $\beta \approx \Phi\left(1.96 - \frac{20\sqrt{n}}{70}\right) < 0.3 \implies 1.96 - \frac{2\sqrt{n}}{7} < -0.5244 \implies n \geq 76$

7. Louis's Expected Gain and Bribes

- $E[X_{\max}] = \int_0^{1000} x \cdot \frac{nx^{n-1}}{1000^n} dx = \frac{1000n}{n+1}$
- $E[\text{Gain}(n)] = \frac{1000n}{n+1} - 5n$
- Maximize: $\frac{d}{dn} E[\text{Gain}] = \frac{1000}{(n+1)^2} - 5 = 0 \implies n+1 = \sqrt{200} \approx 14.14 \implies n = 13$ (gain $\approx \$863.57$)

8. Expected Resell Value

- $\mathbb{E}[U] = 1500$
- $\mathbb{E}[V] = 0(0.3) + (400 + 600 + 650)\frac{7}{30} = 385$
- $\mathbb{E}[\text{Resell}] = 0.3\mathbb{E}[U] + 0.7\mathbb{E}[V] = 0.3(1500) + 0.7(385) = 719.5$